

HONEYCHAIN

Blockchain-Based Honey Traceability & Smart Beekeeping System

*Blockchain Traceability • IoT Hive Monitoring • AI Hive Analysis
QR Consumer Verification • Digital Marketplace for Beekeepers*

Project ID : **Batch 09 (25IT-MP-09)**
Team Members : 1. Sakeena Nivas S (25ITR122)
2. Srinivasan S (25ITR141)
3. Swetha S (25ITR152)
4. Udishkrishna V (25ITR165)
Project Guide : Dr. / Prof. [Faculty Guide Name]
Department of Information Technology



OBJECTIVE

- Reduce counterfeit / adulterated honey through better traceability
- Track honey from hive → beekeeper → processing → customer
- Improve market access for rural beekeepers
- Monitor hive conditions using IoT sensors
- Use AI for hive-health analysis and productivity prediction
- Enable customers to access batch and testing information through QR codes

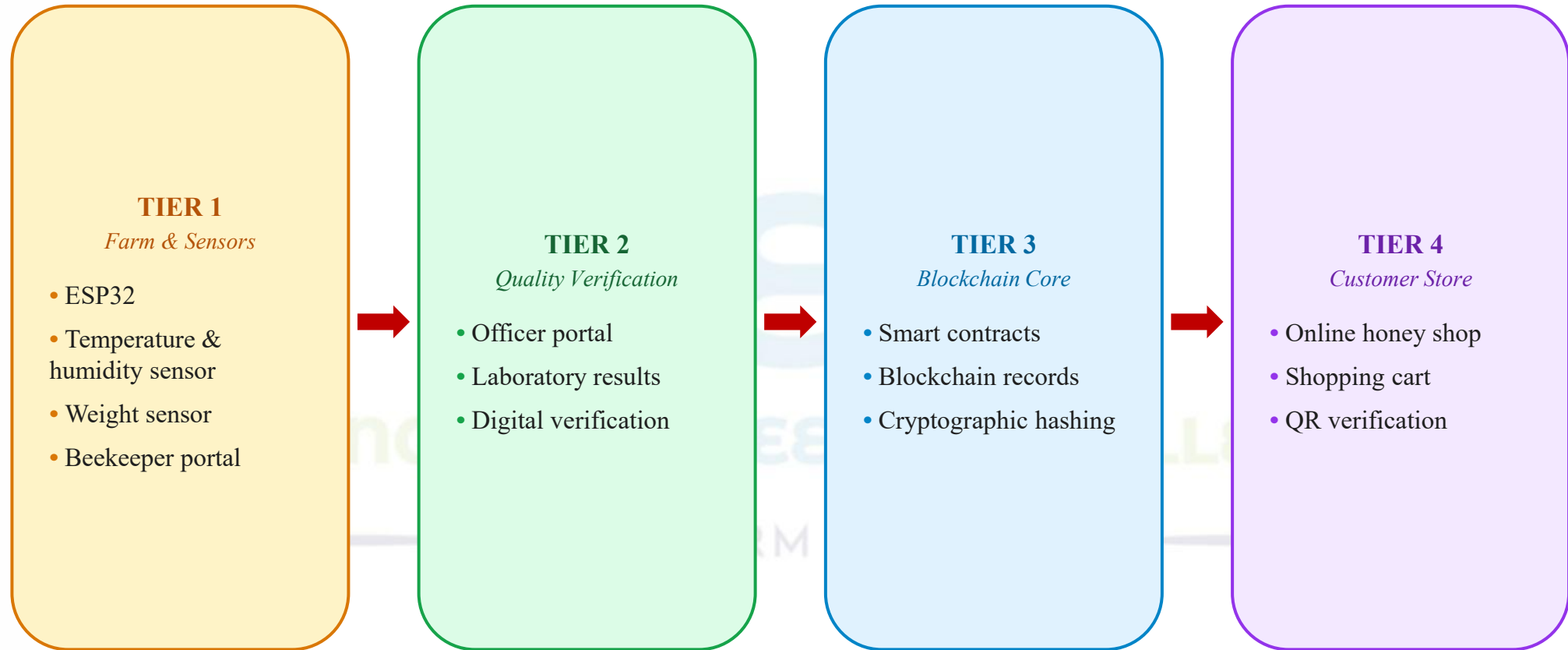
PROBLEM STATEMENT

- Counterfeit / adulterated honey affects consumer trust
- Customers have limited information about honey origin
- Traditional records can be difficult to verify
- Rural beekeepers have limited direct market access
- Manual hive monitoring provides limited real-time information
- Early identification of abnormal hive conditions can be difficult

PROPOSED METHODOLOGY

- **1. Hive Monitoring:** IoT sensors collect temperature, humidity and weight data.
- **2. AI Analysis:** AI analyzes suitable hive data / images for abnormal conditions.
- **3. Honey Harvesting:** Beekeeper records date, quantity, hive and location.
- **4. Quality Testing:** Authorized testing personnel record laboratory test results.
- **5. Blockchain Recording:** Important batch and verification records are stored securely.
- **6. QR Verification:** Customer scans the QR code to access batch information.

BLOCK DIAGRAM / FLOW CHART



System Flow: IoT Sensor Data → Lab Quality Verification → Blockchain Storage → Customer QR Verification

LIST OF MODULES

- **IoT Telemetry Module:** Temperature, humidity & hive weight
- **AI Analytics Module:** Hive health analysis & productivity prediction
- **Beekeeper Portal:** Hive and honey-batch management
- **Quality Officer Portal:** Testing and verification records
- **Blockchain Module:** Secure and traceable batch records
- **QR Code Module:** QR generation and verification
- **Customer Store & Cart:** Product browsing, cart and purchasing

KONGU ENGINEERING COLLEGE
— TRANSFORM YOURSELF —

HARDWARE / SOFTWARE REQUIREMENTS

HARDWARE REQUIREMENTS

- Laptop / Computer
- Smartphone with camera
- ESP32
- DHT22 Sensor
- HX711 + Load Cell

SOFTWARE REQUIREMENTS

- HTML5, CSS3, JavaScript
- Node.js & Express.js
- Solidity & Blockchain platform
- SHA-256 Cryptographic Hashing
- Python & TensorFlow
- VS Code, Git & Google Chrome

MODULES DESCRIPTION

Beekeeper

- Register hives
- Monitor IoT data
- View AI alerts
- Add honey batches

Quality Officer

- Inspect batches
- Enter laboratory results
- Verify batch information
- Approve/reject according to workflow

Customer

- Browse honey
- Manage cart
- Purchase products
- Scan QR and view batch information

Admin

- Manage users
- Manage system
- Monitor blockchain integration

EXPECTED OUTCOME & ADVANTAGES

EXPECTED OUTCOMES

- Transparent honey traceability system
- Digital connection between beekeepers, verification authorities and customers
- Improved market access for rural beekeepers
- Real-time hive monitoring
- AI-based alerts and productivity insights
- Quick QR-based batch verification

KEY ADVANTAGES

-  Tamper-resistant records
-  Easy QR verification
-  Smart hive monitoring
-  AI-based analysis
-  Digital marketplace
-  Better traceability for rural honey production

SDG GOALS

- **SDG 8 – Decent Work & Economic Growth:** Supports rural beekeepers through digital market access and livelihood opportunities.
- **SDG 9 – Industry, Innovation & Infrastructure:** Uses AI, IoT and blockchain for innovation in beekeeping.
- **SDG 12 – Responsible Consumption & Production:** Improves transparency and traceability in honey production.
- **SDG 15 – Life on Land:** Supports better honeybee colony monitoring and sustainable ecosystem management.

REFERENCES

- [1] **KVIC** – Honey Mission and beekeeping support schemes, Ministry of MSME, Govt. of India.
- [2] **FSSAI** – Honey quality, NMR testing protocols and food-safety standards.
- [3] **Satoshi Nakamoto** – Bitcoin: A Peer-to-Peer Electronic Cash System.
- [4] **Ethereum / Vitalik Buterin** – Ethereum blockchain concepts and smart contract architecture.
- [5] **FAO** – Beekeeping and sustainable agriculture resources, Food and Agriculture Organization.
- [6] **IEEE Research Papers** – Blockchain-based food traceability and supply chain verification.
- [7] **Research Literature** – IoT-based hive monitoring and AI-based bee health analysis.